

Girish Yandigeri

PROGRAM 9 – Kubernetes Namespaces & Pods

Commands

`kubectl apply -f namespaces.yaml`

Creates the dev and prod namespaces.

`kubectl get namespaces`

Lists all namespaces in the cluster.

`kubectl apply -f pods.yaml`

Creates frontend and backend pods in both namespaces.

`kubectl get pods -n dev`

Shows all pods running in the dev namespace.

`kubectl get pods -n prod`

Shows all pods running in the prod namespace.

```
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/nor... x girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/nor... x girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/nor... x
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/nor... x$ nano namespaces.yaml
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/nor... x$ kubectl apply -f namespaces.yaml
namespace/dev created
namespace/prod created
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/nor... x$ kubectl get namespaces
NAME      STATUS   AGE
default   Active   7d9h
dev        Active   9s
kube-node-lease   Active   7d9h
kube-public   Active   7d9h
kube-system   Active   7d9h
kubernetes-dashboard   Active   8h
prod        Active   8s
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/nor... x$ kubectl apply -f namespaces.yaml
namespace/dev unchanged
namespace/prod unchanged
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/nor... x$ kubectl get namespaces
NAME      STATUS   AGE
default   Active   7d9h
dev        Active   27s
kube-node-lease   Active   7d9h
kube-public   Active   7d9h
kube-system   Active   7d9h
kubernetes-dashboard   Active   8h
prod        Active   26s
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/nor... x$ nano pods.yaml
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/nor... x$ kubectl apply -f pods.yaml
pod/frontend-pod created
pod/backend-pod created
pod/frontend-pod created
pod/backend-pod created
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/nor... x$ kubectl get pods -n dev
NAME      READY   STATUS    RESTARTS   AGE
backend-pod   0/1     ContainerCreating   0          5s
frontend-pod   0/1     ContainerCreating   0          6s
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/nor... x$ kubectl get pods -n dev -l app=frontend
NAME      READY   STATUS    RESTARTS   AGE
backend-pod   0/1     ContainerCreating   0          6s
frontend-pod   0/1     ContainerCreating   0          6s
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/nor... x$ kubectl get pods -n dev -l app=backend
NAME      READY   STATUS    RESTARTS   AGE
backend-pod   0/1     ContainerCreating   0          9s
frontend-pod   0/1     ContainerCreating   0          9s
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/nor... x$ kubectl get pods -n prod -l app=frontend
NAME      READY   STATUS    RESTARTS   AGE
backend-pod   0/1     ContainerCreating   0          10s
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/nor... x$ kubectl get pods -n prod -l app=backend
NAME      READY   STATUS    RESTARTS   AGE
backend-pod   0/1     ContainerCreating   0          9s
frontend-pod   0/1     ContainerCreating   0          9s
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/nor... x$ kubectl get pods -n prod
NAME      READY   STATUS    RESTARTS   AGE
backend-pod   0/1     ContainerCreating   0          10s
frontend-pod   0/1     ContainerCreating   0          10s
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/nor... x$
```

`kubectl get pods -n dev -l app=frontend`

Displays only frontend pods in the dev namespace.

`kubectl get pods -n dev -l app=backend`

Displays only backend pods in the dev namespace.

`kubectl get pods -n prod -l app=frontend`

Displays only frontend pods in the prod namespace.

```
kubecttl get pods -n prod -l app=backend
```

Displays only backend pods in the prod namespace.

```
kubecttl delete pods --all -n dev
```

Deletes all pods from the dev namespace.

```
kubecttl delete pods --all -n prod
```

Deletes all pods from the prod namespace.

```
kubecttl delete namespace dev
```

Deletes the dev namespace.

```
kubecttl delete namespace prod
```

Deletes the prod namespace.

```
kubecttl get pods --all-namespaces
```

Verifies that no pods remain in any namespace.

namespaces.yaml

```
apiVersion: v1
```

```
kind: Namespace
```

```
metadata:
```

```
  name: dev
```

```
---
```

```
apiVersion: v1
```

```
kind: Namespace
```

```
metadata:
```

```
  name: prod
```

pods.yaml

```
apiVersion: v1
```

```
kind: Pod
```

```
metadata:
```

```
  name: frontend-pod
```

```
  namespace: dev
```

```
  labels:
```

```
    app: frontend
```

```
    env: dev
```

```
spec:
```

```
  containers:
```

```
    - name: frontend
```

```
      image: nginx
```

```
      ports:
```

```
        - containerPort: 80
```

```
---
```

```
apiVersion: v1
```

```
kind: Pod
```

```
metadata:
```

```
name: backend-pod
namespace: dev
labels:
  app: backend
  env: dev
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  namespace: prod
  labels:
    app: frontend
    env: prod
spec:
  containers:
  - name: frontend
    image: nginx
    ports:
    - containerPort: 80
```

```
apiVersion: v1
kind: Pod
metadata:
  name: backend-pod
  namespace: prod
  labels:
    app: backend
    env: prod
spec:
  containers:
  - name: backend
    image: nginx
    ports:
    - containerPort: 80
```